

WINDOWS SETUP GUIDE

Installing Edge GDS Server on Windows 11

A step-by-step guide to installing, configuring, and running the Windows Service build of the Edge GDS Server — the same OPC UA Global Discovery Server as the Docker deployment, packaged for local install and testing on Windows.

Package: EdgeGDSServer-Setup.exe **Requires:** Windows 11, 64-bit, Administrator

Ports: 4840 (OPC UA) · 8443 (Web UI)

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01 Before you start

- Windows 11, 64-bit. (Windows 10 64-bit will very likely work too, but only Windows 11 has been verified.)
- Administrator rights on the machine — the installer registers a Windows Service and adds firewall rules, both of which need elevation.
- Ports **4840** and **8443** free on the machine. If something else is already listening on either, see [Troubleshooting](#).

ABOUT THIS BUILD

This Windows package is an addition alongside the Docker/Linux deployment, not a replacement for it — it's meant for running and testing the GDS directly on a Windows machine,

without Docker. The service installs with **Manual** start and is **never started automatically** by the installer or on reboot — you start it yourself whenever you want to use it.

02 Running the installer

- 1 Right-click `EdgeGDSServer-Setup.exe` and choose **Run as administrator**. Approve the UAC prompt.
- 2 Step through the setup wizard. The default install location is `C:\Program Files\Edge GDS Server\` — there's no reason to change it unless you have a specific one.
- 3 Click **Install**, then **Finish**.

Behind that one click, the installer:

- copies the application files to the install folder,
- registers a Windows Service named `EdgeGDSServer` — **Manual** start,
- opens Windows Firewall for inbound TCP on `4840` and `8443`,
- writes a starter configuration file, `.env`, next to the installed `.exe`, and
- adds Start Menu shortcuts under **Edge GDS Server** (Web admin UI, Open data folder, Uninstall).

Nothing is started yet — that's the next step.

03 Starting the service

Open an elevated Command Prompt or PowerShell and run:

```
:: start it
net start EdgeGDSServer

:: check it's running
sc query EdgeGDSServer
```

Or, without the command line: open `services.msc`, find **Edge GDS Server** in the list, right-click it, and choose **Start**.

The first start takes a few seconds longer than every start after it — this is when the service generates its root CA and its own TLS certificate for the web UI. There's no console window and

no progress bar; the service is ready once `sc query EdgeGDSServer` reports `RUNNING`, or once `https://localhost:8443/` loads in a browser.

IF IT WON'T START

Check `C:\Program Files\Edge GDS Server\edge-gds-server.log` — the service has no console of its own, so every startup message and error goes there instead. See [Troubleshooting](#) for the most common causes.

04 Finding the initial admin password

No password ships with the installer. The first time the service starts with no admin account yet in its database, it creates one named `admin` with a random password, and prints that password — once — to the log file:

```
C:\Program Files\Edge GDS Server\edge-gds-server.log
=====
Created initial GDS admin account: admin
Generated password (shown only this once): Xk29-qLmPz-...
Set GDS_ADMIN_PASSWORD to control this instead.
=====
```

Open that file in Notepad and copy the password before it scrolls further. If you'd rather set your own password up front instead of hunting through a log, add `GDS_ADMIN_PASSWORD=` to the `.env` file (see [Configuration](#)) before the first start.

05 First login

- 1 Open `https://localhost:8443/` in a browser.
- 2 Your browser will warn that the connection isn't private — expected, and covered in the next section. Click through it (in Chrome/Edge: **Advanced** → **Continue to localhost**).
- 3 Sign in as `admin` with the password from the log file.
- 4 You'll land straight on a **Change password** screen — this first login is only good for one use. Pick a real password and continue.

From here you're at the dashboard, the same admin UI as the Docker deployment: Applications, Certificates, Trust List, Push Clients, Settings, Users.

06 Trusting the certificate (optional, recommended)

The web UI's HTTPS certificate is signed by this GDS's own root CA, not a public one, so every browser will flag it as untrustworthy until that CA is added to Windows' trust store. This is a one-time, optional step — the service works fine without it, you'll just keep seeing the warning.

- 1 With the service running, download <https://localhost:8443/trustlist/ca.crt> — this one link works without logging in, since a root CA certificate is meant to be shared openly.
- 2 Double-click the downloaded `gds-root-ca.crt` file.
- 3 Click **Install Certificate...** → **Local Machine** (needs admin) → **Place all certificates in the following store** → **Trusted Root Certification Authorities** → **Finish**.
- 4 Restart your browser. The warning is gone for good.

WHY BOTHER

Trusting the CA once also means any other OPC UA application you point at this GDS — for certificate pull or push — can validate certificates it issues, not just the web UI.

07 Configuration

Settings live in `C:\Program Files\Edge GDS Server\.env` — plain `KEY=value` lines, one per setting. A full reference copy ships alongside it as `.env.example`. The ones worth knowing about:

GDS_HOSTNAME	The name/IP other machines use to reach this GDS. Defaults to <code>localhost</code> — change this to the machine's real IP or hostname if other OPC UA devices need to reach it over the network.
GDS OPCUA_PORT	OPC UA endpoint port. Default <code>4840</code> .
GDS_HTTP_PORT	Web admin UI port. Default <code>8443</code> .
GDS_DATA_DIR	Where the CA key, certificate database, and session secret live. Set by the installer to <code>C:\ProgramData\Edge GDS Server\data</code> — treat this folder like a secrets store.
GDS_ADMIN_PASSWORD	Set this before first start to skip the random-password-in-the-log step above.

CHANGES NEED A RESTART

The service reads `.env` once, at startup. After editing it:

```
net stop EdgeGDSServer
net start EdgeGDSServer
```

Changing `GDS_HOSTNAME` or the ports after the certificates already exist won't reissue them automatically — delete the relevant files under `C:\ProgramData\Edge GDS Server\data\pki\` first if you need fresh ones (this also means a new root CA, so anything that already trusts the old one will need to trust the new one too).

08 Managing the service

TASK	COMMAND
Start	<code>net start EdgeGDSServer</code>
Stop	<code>net stop EdgeGDSServer</code>
Status	<code>sc query EdgeGDSServer</code>
Start / Stop (GUI)	<code>services.msc</code> → Edge GDS Server

Because it's Manual start, the service does **not** come back on its own after a reboot — start it again the same way whenever you want to use it.

Application	<code>C:\Program Files\Edge GDS Server\</code>
Data (CA, database)	<code>C:\ProgramData\Edge GDS Server\data\</code>
Log file	<code>C:\Program Files\Edge GDS Server\edge-gds-server.log</code>

09 Uninstalling

Use **Settings** → **Apps**, or the **Uninstall Edge GDS Server** shortcut in the Start Menu group. It stops the service, unregisters it, and removes the two firewall rules.

YOUR DATA SURVIVES

`C:\ProgramData\Edge GDS Server\` — the root CA's private key and every issued certificate — is deliberately left behind. If you're done with it for good, delete that folder yourself; if you're reinstalling later, your CA and certificate history will still be there.

10 Troubleshooting

The service won't start

Check `edge-gds-server.log` first — it's the only place startup errors go. The most common cause is a port already in use:

```
netstat -ano | findstr ":4840"  
netstat -ano | findstr ":8443"
```

If either turns up a process, either stop it or change `GDS_OPCUA_PORT` / `GDS_HTTP_PORT` in `.env` (and update the matching firewall rule).

A browser or OPC UA client can't reach it from another machine

The installer opens the firewall on the **Domain** and **Private** network profiles by default. If Windows classified your network as **Public**, add the rule there too, or re-run:

```
netsh advfirewall firewall add rule name="Edge GDS Server (Web UI)" dir=in action=allow proto=TCP port=4840  
netsh advfirewall firewall add rule name="Edge GDS Server (OPC UA)" dir=in action=allow proto=TCP port=8443
```

Browser still shows a certificate warning after trusting the CA

Make sure the certificate was installed into the **Local Machine** store (not **Current User**) under **Trusted Root Certification Authorities**, and fully restart the browser — not just the tab.

Forgot the admin password after changing it

Log in with another admin account if one exists, and reset it from the **Users** page. With no other admin account, there's no self-service recovery — this is the same trade-off the Docker deployment makes, since the whole point is that the GDS never has a way to hand out a certificate authority's credentials to just anyone who asks.

Edge GDS Server — Windows Service build. This package installs alongside, and does not replace, the Docker/Linux deployment.